

Computational Properties and Goodness-of-Fit of the OLS Sample Regression Equations

- State and prove the **five computational properties** of the **OLS SRE**

$$\begin{aligned} Y_i &= \hat{\beta}_0 + \hat{\beta}_1 X_i + \hat{u}_i \\ &= \hat{Y}_i + \hat{u}_i \end{aligned} \quad (i = 1, \dots, N) \quad (1)$$

- Derive and interpret the **OLS decomposition equation**, which looks like this:

$$\sum_{i=1}^N y_i^2 = \sum_{i=1}^N \hat{y}_i^2 + \sum_{i=1}^N \hat{u}_i^2 \quad (5.1)$$

or

$$\text{TSS} = \text{ESS} + \text{RSS} \quad (5.2)$$

Computational Properties and Goodness-of-Fit of the OLS Sample Regression Equations

- Define and interpret the goodness-of-fit measure called R^2 (R-squared), which is defined as

$$R^2 \equiv \frac{\sum_i \hat{y}_i^2}{\sum_i y_i^2} = 1 - \frac{\sum_i \hat{u}_i^2}{\sum_i y_i^2}$$

or

$$R^2 \equiv \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS}$$

Starting Point

The **OLS sample regression equation (OLS-SRE)** is

$$Y_i = \hat{\beta}_0 + \hat{\beta}_1 X_i + \hat{u}_i = \hat{Y}_i + \hat{u}_i \quad (i = 1, \dots, N) \quad (1)$$

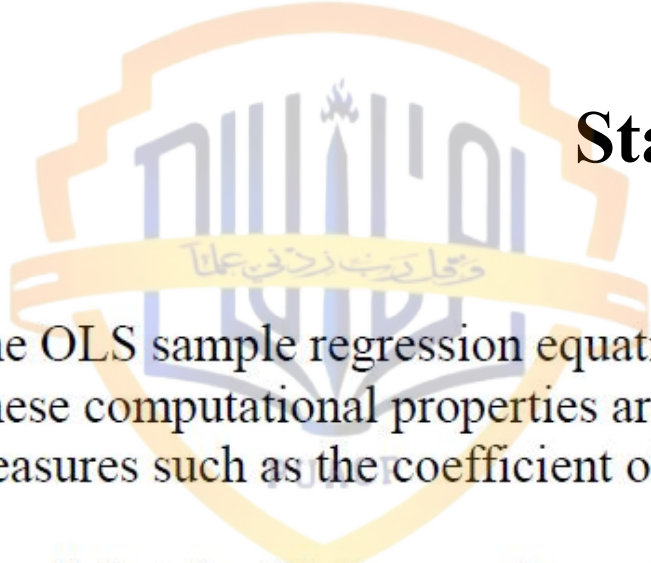
where

$\hat{\beta}_0$ = the OLS estimate of the *intercept coefficient* β_0 ;

$\hat{\beta}_1$ = the OLS estimate of the *slope coefficient* β_1 ;

$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$ = the i-th estimated (or predicted) value of $E(Y_i | X_i) = \beta_0 + \beta_1 X_i$, and is called the **OLS sample regression function** (or **OLS-SRF**);

$\hat{u}_i = Y_i - \hat{Y}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i$ = the i-th **OLS residual**.



Starting Point

The OLS sample regression equation (1) exhibits *five computational properties*. These computational properties are necessary for developing goodness-of-fit measures such as the coefficient of determination, R^2 .

Recall that the **OLS normal equations** for the simple (two-variable) linear regression model are:

$$N\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^N X_i = \sum_{i=1}^N Y_i \quad (\text{N1})$$

$$\hat{\beta}_0 \sum_{i=1}^N X_i + \hat{\beta}_1 \sum_{i=1}^N X_i^2 = \sum_{i=1}^N X_i Y_i \quad (\text{N2})$$

Computational Properties of the OLS SRE

PROPERTY 1: The *OLS sample regression equation* passes through the *point of sample means* $(\bar{Y}, \bar{X})$, where

$\bar{Y} = \sum_{i=1}^N Y_i / N$ is the sample mean value of Y; and

$\bar{X} = \sum_{i=1}^N X_i / N$ is the sample mean value of X.

That is,

$$\bar{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{X} \quad \dots (C1)$$

Computational Properties of the OLS SRE

□ **Proof of (C1):** Follows from the first OLS normal equation (N1)

$$\sum_i Y_i = N\hat{\beta}_0 + \hat{\beta}_1 \sum_i X_i. \quad (\text{N1})$$

Dividing both sides of equation (N1) by N yields

$$\frac{\sum_i Y_i}{N} = \hat{\beta}_0 + \hat{\beta}_1 \frac{\sum_i X_i}{N}$$

or, using the definitions of $\bar{Y}$ and $\bar{X}$,

$$\bar{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{X}. \quad \dots (\text{C1})$$

Computational Properties of the OLS SRE

PROPERTY 2: The sample mean of the *estimated* Y_i 's (the $\hat{Y}_i$'s) *equals* the sample mean of the *observed* Y_i 's; or the sum of the *estimated* Y_i 's (the $\hat{Y}_i$'s) *equals* the sum of the *observed* Y_i 's.

$$\bar{\hat{Y}} = \bar{Y} \quad \text{where } \bar{\hat{Y}} \equiv \sum_i \hat{Y}_i / N \text{ and } \bar{Y} = \sum_i Y_i / N \quad \dots \text{(C2)}$$

or

$$\sum_{i=1}^N \hat{Y}_i = \sum_{i=1}^N Y_i \quad \text{sum of } \textit{estimated} Y_i\text{'s, (the } \hat{Y}_i\text{'s)} = \text{sum of } \textit{observed} Y_i\text{'s.}$$

Computational Properties of the OLS SRE

(1) The estimated values of Y_i are given by

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i.$$

(2) Substitute for $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$ in the above expression for $\hat{Y}_i$:

$$\begin{aligned}\hat{Y}_i &= \bar{Y} - \hat{\beta}_1 \bar{X} + \hat{\beta}_1 X_i \\ &= \bar{Y} + \hat{\beta}_1 (X_i - \bar{X})\end{aligned}$$

(3) Now sum both sides over $i = 1, \dots, N$:

$$\begin{aligned}\sum_{i=1}^N \hat{Y}_i &= N\bar{Y} + \hat{\beta}_1 \left(\sum_{i=1}^N X_i - N\bar{X} \right) \\ &= N\bar{Y} + \hat{\beta}_1 (N\bar{X} - N\bar{X}), && \text{since } \sum_{i=1}^N X_i = N\bar{X} \\ &= N\bar{Y}, && \text{since } (N\bar{X} - N\bar{X}) = 0.\end{aligned}$$

(4) Finally, dividing by N , we get

$$\frac{\sum_{i=1}^N \hat{Y}_i}{N} = \bar{Y} \quad \Rightarrow \quad \overline{\hat{Y}} = \bar{Y} \quad \Rightarrow \quad \sum_{i=1}^N \hat{Y}_i = \sum_{i=1}^N Y_i.$$

- **Implication of Property C2:** The OLS-SRF $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$ can be written in **deviation-from-means form** as

$$\hat{y}_i = \hat{\beta}_1 x_i \quad \text{where } \hat{y}_i \equiv \hat{Y}_i - \bar{Y} \text{ and } x_i \equiv X_i - \bar{X}.$$

Proof:

- (1) From line (2) of the proof of Property (C2) above,

$$\hat{Y}_i = \bar{Y} + \hat{\beta}_1 (X_i - \bar{X}).$$

- (2) Subtract $\bar{Y}$ from both sides of the above equation to get

$$(\hat{Y}_i - \bar{Y}) = \hat{\beta}_1 (X_i - \bar{X}),$$

which is simply

$$\hat{y}_i = \hat{\beta}_1 x_i$$

where by definition $\hat{y}_i \equiv \hat{Y}_i - \bar{Y}$ and $x_i \equiv X_i - \bar{X}$. □



Computational Properties of the OLS SRE

PROPERTY 3: The *sample mean* of the OLS residuals $\hat{u}_i$ equals *zero*, or the *sum* of the OLS residuals $\hat{u}_i$ equals *zero*.

$$\bar{u} = \sum_{i=1}^N \hat{u}_i / N = \frac{\sum_{i=1}^N \hat{u}_i}{N} = 0 \quad \text{or} \quad \sum_{i=1}^N \hat{u}_i = 0. \quad \dots (C3)$$

Computational Properties of the OLS SRE

□ **Proof of (C3):** Involves demonstrating that $\sum_i \hat{u}_i = 0$.

(1) From the first normal equation (N1), we have

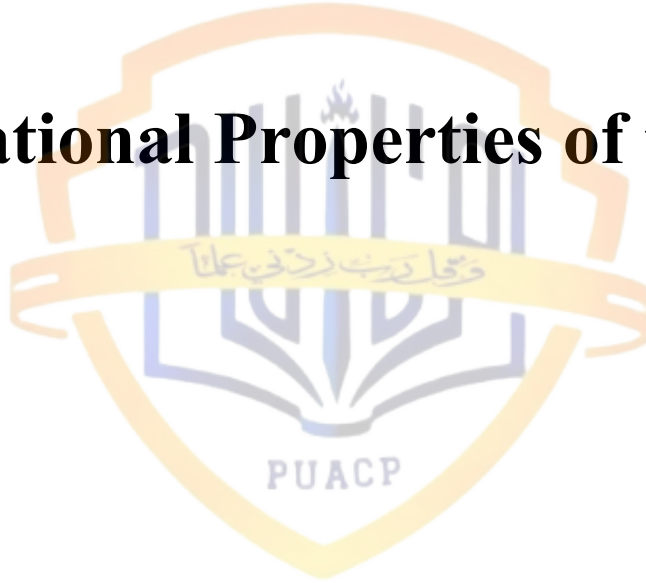
$$\begin{aligned} -2 \sum_{i=1}^N (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) &= 0 \\ \sum_{i=1}^N (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) &= 0. \end{aligned}$$

(2) But $(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = \hat{u}_i$ by definition, so that equation (N1) implies that

$$\sum_{i=1}^N \hat{u}_i = 0 \quad \text{and hence that} \quad \bar{u} = \sum_{i=1}^N \hat{u}_i / N = \frac{\sum_i \hat{u}_i}{N} = 0. \quad \square$$

NOTE: Properties 1-3 depend on their being an *intercept coefficient in the population regression function*. The following two properties do not require an intercept in the regression function.

Computational Properties of the OLS SRE



PROPERTY 4: The OLS residuals $\hat{u}_i$ are *uncorrelated with the sample values* of X , the X_i ; i.e.,

$$\sum_{i=1}^N X_i \hat{u}_i = 0 .$$

... (C4)

Computational Properties of the OLS SRE

□ **Proof of (C4):** Is based on the second OLS normal equation (N2):

$$\sum_{i=1}^N X_i Y_i = \hat{\beta}_0 \sum_{i=1}^N X_i + \hat{\beta}_1 \sum_{i=1}^N X_i^2 . \quad (\text{N2})$$

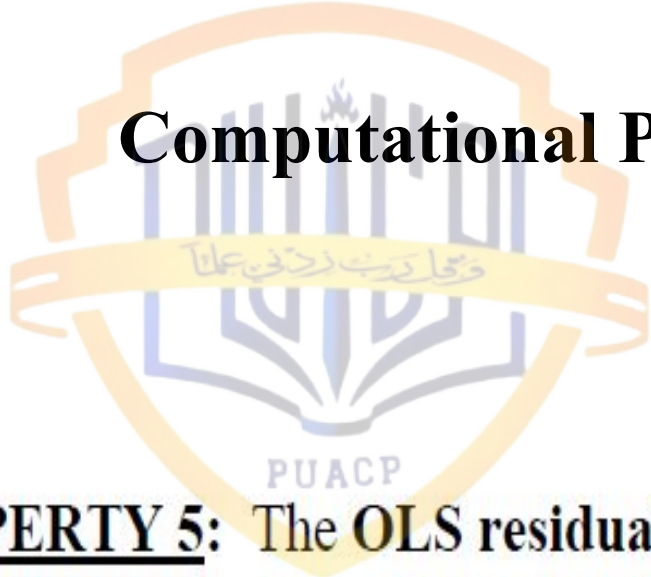
(1) Since $\hat{u}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i$, we can pre-multiply by X_i to obtain

$$X_i \hat{u}_i = X_i Y_i - \hat{\beta}_0 X_i - \hat{\beta}_1 X_i^2 .$$

(2) Summing over $i = 1, \dots, N$, we get

$$\begin{aligned} \sum_{i=1}^N X_i \hat{u}_i &= \sum_{i=1}^N X_i Y_i - \hat{\beta}_0 \sum_{i=1}^N X_i - \hat{\beta}_1 \sum_{i=1}^N X_i^2 \\ &= 0 \end{aligned}$$

by normal equation (N2).



Computational Properties of the OLS SRE

PROPERTY 5: The OLS residuals $\hat{u}_i$ are *uncorrelated* with the *estimated* or *predicted* values of Y_i , the $\hat{Y}_i$; i.e.,

$$\sum_{i=1}^N \hat{Y}_i \hat{u}_i = 0 .$$

... (C5)

Computational Properties of the OLS SRE

□ **Proof of (C5):** Makes use of properties (C3) and (C4) above.

(1) Since $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$, we can post-multiply by $\hat{u}_i$ to obtain

$$\hat{Y}_i \hat{u}_i = \hat{\beta}_0 \hat{u}_i + \hat{\beta}_1 X_i \hat{u}_i.$$

(2) Summing over $i = 1, \dots, N$, we get

$$\sum_{i=1}^N \hat{Y}_i \hat{u}_i = \hat{\beta}_0 \sum_{i=1}^N \hat{u}_i + \hat{\beta}_1 \sum_{i=1}^N X_i \hat{u}_i.$$

(3) But $\sum_{i=1}^N \hat{u}_i = 0$ by (C3) and $\sum_{i=1}^N X_i \hat{u}_i = 0$ by (C4), so that

$$\sum_{i=1}^N \hat{Y}_i \hat{u}_i = 0.$$

Goodness-of-Fit of the OLS-SRE

Our objective now is to derive a measure of how well the OLS-SRE fits the sample data.

- The measure of goodness-of-fit we use is called the *coefficient of determination*, which is conventionally denoted as R^2 .
- The R^2 provides a measure of how well the OLS-SRE explains, or accounts for, the observed sample variation of the regressand Y , where

$$\text{sample variation of } Y \equiv \sum_{i=1}^N (Y_i - \bar{Y})^2 = \sum_{i=1}^N y_i^2 .$$

- The derivation of the R^2 for an OLS-SRE is based on the **OLS decomposition equation** for the sample variation of Y .

The OLS Decomposition Equation

3.1 Derivation of the OLS Decomposition Equation

1. For each sample observation i , the OLS-SRE is written as

$$Y_i = \hat{\beta}_0 + \hat{\beta}_1 X_i + \hat{u}_i$$

or

(1)

$$Y_i = \hat{Y}_i + \hat{u}_i \quad \text{where } \hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i \quad (i = 1, \dots, N)$$

2. Subtract the sample mean of the Y_i values, $\bar{Y}$, from both sides of equation (1):

$$Y_i - \bar{Y} = \hat{Y}_i - \bar{Y} + \hat{u}_i$$

or, in deviation-from-means form,

$$y_i = \hat{y}_i + \hat{u}_i \quad (2)$$

where $y_i \equiv Y_i - \bar{Y}$, $\hat{y}_i \equiv \hat{Y}_i - \bar{Y} = \hat{\beta}_1 x_i$, and $x_i \equiv X_i - \bar{X}$.

The OLS Decomposition Equation

where $y_i \equiv Y_i - \bar{Y}$, $\hat{y}_i \equiv \hat{Y}_i - \bar{Y} = \hat{\beta}_1 x_i$, and $x_i \equiv X_i - \bar{X}$.

3. Next, square both sides of equation (2):

$$\begin{aligned} y_i^2 &= (\hat{y}_i + \hat{u}_i)^2 \\ &= \hat{y}_i^2 + \hat{u}_i^2 + 2\hat{y}_i\hat{u}_i \end{aligned} \tag{3}$$

4. Now sum both sides of equation (3) over $i = 1, \dots, N$:

$$\sum_{i=1}^N y_i^2 = \sum_{i=1}^N \hat{y}_i^2 + \sum_{i=1}^N \hat{u}_i^2 + 2 \sum_{i=1}^N \hat{y}_i \hat{u}_i \tag{4}$$

The OLS Decomposition Equation

5. But the last term on the right-hand side of equation (4) equals zero:

$$\begin{aligned}\sum_{i=1}^N \hat{y}_i \hat{u}_i &= \sum_{i=1}^N (\hat{Y}_i - \bar{Y}) \hat{u}_i \\ &= \sum_{i=1}^N \hat{Y}_i \hat{u}_i - \bar{Y} \sum_{i=1}^N \hat{u}_i \\ &= 0 \quad \text{since } \sum_{i=1}^N \hat{Y}_i \hat{u}_i = 0 \text{ by (C5) and } \sum_{i=1}^N \hat{u}_i = 0 \text{ by (C3).}\end{aligned}$$

6. Therefore, setting $\sum_i \hat{y}_i \hat{u}_i = 0$ in equation (4) gives the result that

$$\sum_{i=1}^N y_i^2 = \sum_{i=1}^N \hat{y}_i^2 + \sum_{i=1}^N \hat{u}_i^2. \quad (5)$$

□ **Result:** Equation (5) is the OLS decomposition equation for OLS-SRE (1).

Interpretation of the OLS Decomposition Equation

Equation (5) is the OLS decomposition equation for OLS-SRE (1):

$$\sum_{i=1}^N y_i^2 = \sum_{i=1}^N \hat{y}_i^2 + \sum_{i=1}^N \hat{u}_i^2 . \quad (5)$$

Each of the three terms in equation (5) are defined as follows:

- (1) $\sum_{i=1}^N y_i^2 \equiv \text{TSS} \equiv$ the Total Sum of Squares
- = the total sum of squares of the observed sample values of Y about their sample mean $\bar{Y}$
 - = the total sample variation of the observed Y_i values.

Interpretation of the OLS Decomposition Equation

- (2) $\sum_{i=1}^N \hat{y}_i^2 \equiv \text{ESS} \equiv$ the Explained Sum of Squares
= the sum of squares of the estimated or predicted values of Y, the $\hat{Y}_i$, about their sample mean $\bar{Y}$
= the sum of squares explained by the sample regression function, i.e., by the regressor X.
- (3) $\sum_{i=1}^N \hat{u}_i^2 \equiv \text{RSS} \equiv$ the Residual Sum of Squares
= the sum of squares of the OLS residuals $\hat{u}_i$
= the unexplained variation of the observed sample values Y_i of the regressand Y around the sample regression line

Interpretation of the OLS Decomposition Equation

Using these definitions, the OLS decomposition equation

$$\sum_{i=1}^N y_i^2 = \sum_{i=1}^N \hat{y}_i^2 + \sum_{i=1}^N \hat{u}_i^2 . \quad (5.1)$$

can be re-written as

$$\text{TSS} = \text{ESS} + \text{RSS} \quad (5.2)$$

Equation (5.1) or (5.2) -- the OLS decomposition equation -- decomposes the **sample variation of the regressand Y** into *two additive components*:

- (1) one component, $\text{ESS} = \sum_{i=1}^N \hat{y}_i^2$, is attributable to, or explained by, the sample regression function $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$;
- (2) a second component, $\text{RSS} = \sum_{i=1}^N \hat{u}_i^2$, is attributable to the OLS residuals $\hat{u}_i$ representing unknown random factors that influence the observed Y_i values.

An Unbiased Estimator of the Error Variance

The Residual Sum of Squares (RSS) in the OLS decomposition equation can be used to construct an *unbiased estimator* of the *unknown error variance* σ^2 .

- **Question:** Why do we need an estimator of the error variance σ^2 ?
- **Answer:** We need an estimator of the error variance σ^2 so that we can obtain estimators of the variances of the OLS coefficient estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ which as we have seen are given by the formulas

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum_i x_i^2} = \frac{\sigma^2}{\sum_i (X_i - \bar{X})^2}$$

$$\text{Var}(\hat{\beta}_0) = \frac{\sigma^2 \sum_i X_i^2}{N \sum_i x_i^2} = \frac{\sigma^2 \sum_i X_i^2}{N \sum_i (X_i - \bar{X})^2}.$$

- **Result:** An *unbiased estimator* of the **error variance** σ^2 is given by the formula

$$\hat{\sigma}^2 = \frac{\sum_i \hat{u}_i^2}{(N-2)} = \frac{\text{RSS}}{(N-2)}, \quad \hat{u}_i = Y_i - \hat{Y}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i \quad (i = 1, \dots, N)$$

where 2 is the number of regression coefficients estimated, and $N-2$ is the degrees of freedom for RSS.

- **Explanation:** $\hat{\sigma}^2$ is an unbiased estimator of the error variance because it can be shown that

$$E(\text{RSS}) = E\left(\sum_i \hat{u}_i^2\right) = (N-2)\sigma^2.$$

Therefore

$$E(\hat{\sigma}^2) = E\left(\frac{\sum_i \hat{u}_i^2}{(N-2)}\right) = \frac{E\left(\sum_i \hat{u}_i^2\right)}{(N-2)} = \frac{(N-2)\sigma^2}{(N-2)} = \sigma^2.$$

Summary: $\hat{\sigma}^2$ is an *unbiased estimator* of the error variance σ^2 :

$$E(\hat{\sigma}^2) = \sigma^2 \quad \text{because} \quad E(\text{RSS}) = E\left(\sum_i \hat{u}_i^2\right) = (N-2)\sigma^2.$$

Unbiased Estimators of the Variances of the OLS Coefficient Estimates

Unbiased estimators of $\text{Var}(\hat{\beta}_1)$ and $\text{Var}(\hat{\beta}_0)$ are obtained by simply replacing the unknown σ^2 with its unbiased estimator $\hat{\sigma}^2$ in the formulas for $\text{Var}(\hat{\beta}_1)$ and $\text{Var}(\hat{\beta}_0)$.

- The **unbiased estimator of $\text{Var}(\hat{\beta}_1)$** , the variance of $\hat{\beta}_1$, is therefore

$$\hat{\text{Var}}(\hat{\beta}_1) = \frac{\hat{\sigma}^2}{\sum_i x_i^2} = \frac{\hat{\sigma}^2}{\sum_i (X_i - \bar{X})^2}.$$

- The **unbiased estimator of $\text{Var}(\hat{\beta}_0)$** , the variance of $\hat{\beta}_0$, is therefore

$$\hat{\text{Var}}(\hat{\beta}_0) = \frac{\hat{\sigma}^2 \sum_i X_i^2}{N \sum_i x_i^2} = \frac{\hat{\sigma}^2 \sum_i X_i^2}{N \sum_i (X_i - \bar{X})^2}.$$

The Coefficient of Determination, R²

4.1 Definition of R²

1. Start with the OLS decomposition equation (5.1) or (5.2):

$$\sum_{i=1}^N y_i^2 = \sum_{i=1}^N \hat{y}_i^2 + \sum_{i=1}^N \hat{u}_i^2 \quad (5.1)$$

$$\text{TSS} = \text{ESS} + \text{RSS} \quad (5.2)$$

2. Divide both sides of the OLS decomposition equation (5.1) or (5.2) by TSS =

$$\sum_{i=1}^N y_i^2 :$$

The Coefficient of Determination, R^2

$$1 = \frac{\sum_i \hat{y}_i^2}{\sum_i y_i^2} + \frac{\sum_i \hat{u}_i^2}{\sum_i y_i^2} \quad (6.1)$$

or

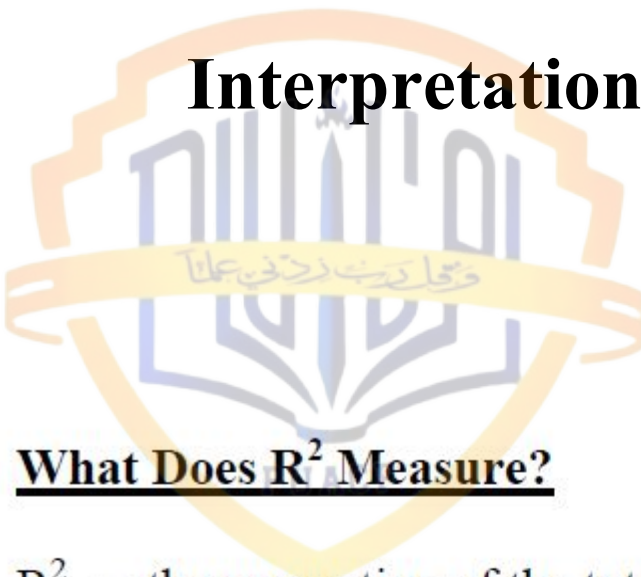
$$1 = \frac{ESS}{TSS} + \frac{RSS}{TSS} \quad (6.2)$$

3. The **coefficient of determination R^2** is defined as:

$$R^2 \equiv \frac{\sum_i \hat{y}_i^2}{\sum_i y_i^2} = 1 - \frac{\sum_i \hat{u}_i^2}{\sum_i y_i^2} \quad \text{from equation (6.1)}$$

or

$$R^2 \equiv \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS} \quad \text{from equation (6.2)}$$



Interpretation of R^2 : The Values of R^2

□ What Does R^2 Measure?

R^2 = the proportion of the total sample variation of the dependent variable Y that is explained by the sample regression function, i.e., by the values of the regressor X.

□ The Values of R^2

R^2 values lie in the closed unit interval $[0, 1]$; i.e., $0 \leq R^2 \leq 1$.

□ **Interpreting the Values of R^2**

- **Rule 1:** The *closer is the value of R^2 to 1*, the *better the goodness-of-fit* of the OLS-SRE to the sample data.
- The upper limiting value $R^2 = 1$ corresponds to a **perfect fit** of the OLS-SRE to the sample data.

$$R^2 = 1 \Rightarrow \frac{ESS}{TSS} = 1 \Rightarrow ESS = TSS \Rightarrow RSS = \sum_i \hat{u}_i^2 = 0.$$

- But since $\hat{u}_i^2 \geq 0$ for all i , $RSS = \sum_i \hat{u}_i^2 = 0$ if and only if

$$\hat{u}_i = 0 \quad \forall \quad i = 1, \dots, N.$$

- Therefore, a perfect fit of the OLS-SRE means that

$$\hat{u}_i = 0 \quad \forall \quad i = 1, \dots, N \quad \text{or} \quad Y_i = \hat{Y}_i \quad \forall \quad i = 1, \dots, N.$$

- **Rule 2:** The *closer* is the value of R^2 to 0, the *worse* the goodness-of-fit of the OLS-SRE to the sample data.
- The lower limiting value $R^2 = 0$ corresponds to the **worst possible fit** of the OLS-SRE to the sample data.

$$R^2 = 0 \Rightarrow \frac{ESS}{TSS} = 0 \Rightarrow ESS = 0 \Rightarrow TSS = RSS.$$

- But $ESS = 0$ if and only if $\hat{\beta}_1 = 0$:

$$ESS = 0 \Rightarrow \sum_{i=1}^N \hat{y}_i^2 = \hat{\beta}_1^2 \sum_{i=1}^N x_i^2 = 0 \Rightarrow \hat{\beta}_1 = 0.$$

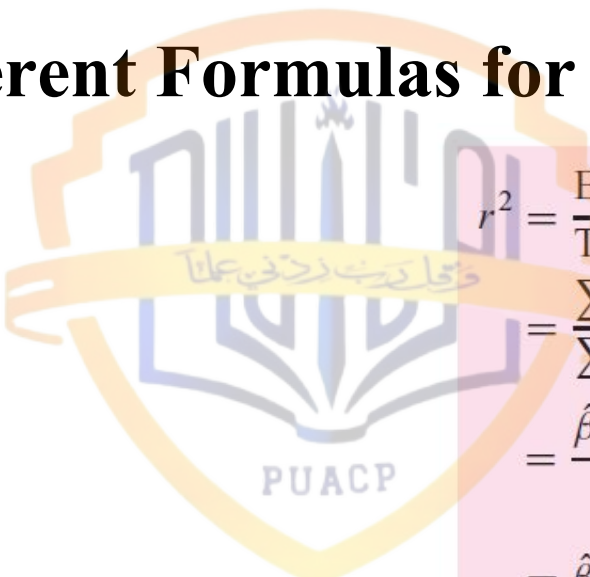
- Finally, since $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$ and $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$, it follows that $R^2 = 0$ means that

$$\hat{Y}_i = \hat{\beta}_0 = \bar{Y} \quad \forall i = 1, \dots, N.$$

The reason is that

$$\begin{aligned} \hat{\beta}_1 = 0 &\Rightarrow \hat{Y}_i = \hat{\beta}_0 && \text{since } \hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i \\ &\Rightarrow \hat{Y}_i = \hat{\beta}_0 = \bar{Y} && \text{since } \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}. \end{aligned}$$

Different Formulas for the Coefficient of Determination, R^2


$$\begin{aligned} r^2 &= \frac{ESS}{TSS} \\ &= \frac{\sum \hat{y}_i^2}{\sum y_i^2} \\ &= \frac{\hat{\beta}_2^2 \sum x_i^2}{\sum y_i^2} \\ &= \hat{\beta}_2^2 \left(\frac{\sum x_i^2}{\sum y_i^2} \right) \end{aligned} \quad (3.5.6)$$

If we divide the numerator and the denominator of Equation 3.5.6 by the sample size n (or $n - 1$ if the sample size is small), we obtain

$$r^2 = \hat{\beta}_2^2 \left(\frac{S_x^2}{S_y^2} \right) \quad (3.5.7)$$

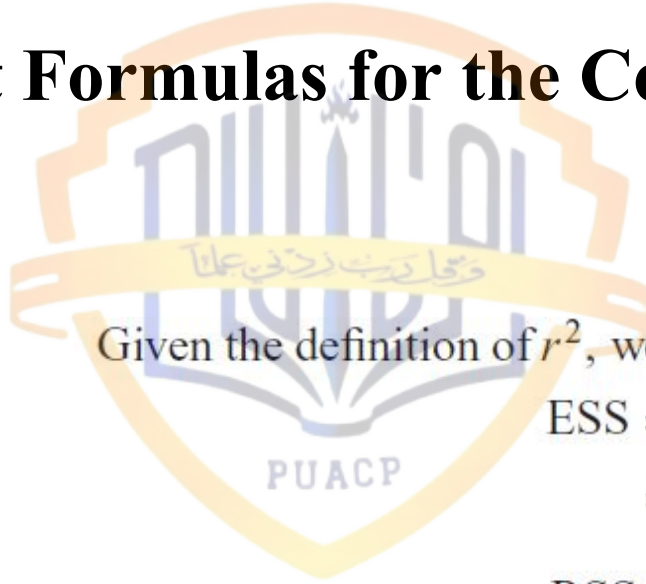
where S_y^2 and S_x^2 are the sample variances of Y and X , respectively.

Since $\hat{\beta}_2 = \sum x_i y_i / \sum x_i^2$, Eq. (3.5.6) can also be expressed as

$$r^2 = \frac{(\sum x_i y_i)^2}{\sum x_i^2 \sum y_i^2} \quad (3.5.8)$$

an expression that may be computationally easy to obtain.

Different Formulas for the Coefficient of Determination, R^2



Given the definition of r^2 , we can express ESS and RSS discussed earlier as follows:

$$\begin{aligned} \text{ESS} &= r^2 \cdot \text{TSS} \\ &= r^2 \sum y_i^2 \end{aligned} \quad (3.5.9)$$

$$\begin{aligned} \text{RSS} &= \text{TSS} - \text{ESS} \\ &= \text{TSS}(1 - \text{ESS}/\text{TSS}) \\ &= \sum y_i^2 \cdot (1 - r^2) \end{aligned} \quad (3.5.10)$$

Therefore, we can write

$$\begin{aligned} \text{TSS} &= \text{ESS} + \text{RSS} \\ \sum y_i^2 &= r^2 \sum y_i^2 + (1 - r^2) \sum y_i^2 \end{aligned} \quad (3.5.11)$$

an expression that we will find very useful later.

Different Formulas for the Coefficient of Determination, R^2

$$r = \frac{\sum x_i y_i}{\sqrt{(\sum x_i^2)(\sum y_i^2)}} \\ = \frac{n \sum X_i Y_i - (\sum X_i)(\sum Y_i)}{\sqrt{[n \sum X_i^2 - (\sum X_i)^2][n \sum Y_i^2 - (\sum Y_i)^2]}}$$

(3.5.13)

$$r^2 = \frac{[\sum (Y_i - \bar{Y})(\hat{Y}_i - \bar{Y})]^2}{\sum (Y_i - \bar{Y})^2 \sum (\hat{Y}_i - \bar{Y})^2}$$

$$r^2 = \frac{(\sum y_i \hat{y}_i)^2}{(\sum y_i^2)(\sum \hat{y}_i^2)}$$